Wenzhe Zhu

wzhzhu@ustc.edu.cn · · (+86) 18855726088 · Homepage

Research Profile

I am a postdoctoral fellow at the University of Science and Technology of China. My research focuses on building scalable and resource-efficient storage, memory, and data-serving systems for data-intensive computing and emerging AI workloads. My work spans distributed storage systems, key-value stores, disaggregated memory, and checkpointing systems for large language model training. My research has appeared in top-tier systems and data management venues, including *SIGMETRICS*, *OSDI*, *FAST*, *ICDE*, *ICS*, *ICPP*, and *SRDS*.

Education

University of Science and Technology of China, Hefei, China *2018 – 2025*

Ph.D. in Computer Science
Thesis: Performance and Overhead Optimization of Log-Structured Cloud Storage Systems
Advisors: Prof. Yinlong Xu and Prof. Yongkun Li

Hunan University, Changsha, China *2014 – 2018*

B.Eng. in Software Engineering

Research Experience & Projects

Distributed and Cloud Storage Systems

- **Traffic scheduling for cloud block storage** *2020 – 2022*
  Studied traffic imbalance and tail-latency amplification in large-scale cloud block storage systems under bursty workloads. Developed differentiated forwarding mechanisms to balance traffic across storage nodes and reduce tail latency. Published in *ACM SIGMETRICS 2023*.
- **Data passing for stateful serverless analytics** *2022 – 2023*
  Identified intermediate data movement as a major performance and cost bottleneck in I/O-intensive stateful serverless analytics. Designed a high-performance, cost-efficient data-passing mechanism that reduces data-movement overhead and improves end-to-end execution efficiency. Published in *FAST 2024*.
- **Load balancing for distributed key-value stores** *2023 – 2025*
  Characterized cross-machine load imbalance in distributed key-value stores caused by heterogeneous CPU and disk-I/O demands under mixed workloads. Designed a cooperative CPU–I/O scheduling mechanism that coordinates compute and storage resources across nodes, improving load balance, throughput, and tail latency. Published in *ICDE 2026*.
- **Efficient erasure-coded key-value storage** *2025 – 2026*
  Led the study of update and repair overheads in erasure-coded key-value stores under dynamic workloads. Developed dynamic decoupling mechanisms that bypass update and repair penalties while preserving storage efficiency and fault tolerance. Accepted to appear in *SRDS 2026*.

Storage Optimizations for Data-Intensive Applications

- **Incremental checkpointing for LLM training** *2024 – 2025*
  Led the characterization of storage and I/O bottlenecks introduced by frequent checkpointing in large language model training. Designed and implemented a fast, space-efficient incremental checkpointing mechanism that reduces checkpoint storage overhead and checkpointing time. Published in *ICPP 2025*.
- **Resource-efficient key-value engine optimization** *2025 – Present*
  Led a study of cache inefficiencies in LSM-tree-based key-value engines under constrained memory resources. Developed cooperative metadata–data cache management to improve cache utilization and read performance by jointly managing index/filter metadata and data blocks. Accepted to appear in *ICS 2026*.

Currently extending this work toward tiered caching architectures that move beyond monolithic cache designs and optimize cache placement across heterogeneous memory and storage resources.

- **Resource-efficient vector indexing and caching** *2025 – Present*
  Leading the design of memory- and scalability-efficient vector indexing and caching mechanisms. Exploring topology-aware shard partitioning mechanisms that reduce scatter-gather query overhead in distributed vector databases, and workload-aware caching for vector indexes to improve lookup and retrieval latency under resource constraints. Ongoing work.

Disaggregated Memory Systems

- **Fine-grained disaggregated memory management** *2024 – 2025*
  Studied the trade-off between allocation overhead and memory waste in fine-grained disaggregated memory systems. Designed a memory management framework that reduces allocation overhead while preserving fine-grained memory efficiency. Published in *OSDI 2025*.
- **Elastic and low-latency remote swap for disaggregated memory** *2025 – Present*
  Investigated RDMA-based remote swap for transparent memory expansion in disaggregated memory architectures. Designed elastic fine-grained memory slot allocation to reduce memory stranding caused by static peak-demand provisioning, and redesigned swap metadata management with reference-count arrays and hierarchical queues to achieve $O(1)$ swap entry allocation. Ongoing work.

Professional Experience

University of Science and Technology of China, Hefei, China *Aug. 2025 – now*

Postdoctoral Fellow

Working with Prof. Yongkun Li.

Alibaba Cloud, Beijing, China *Nov. 2019 – Jun. 2021*

Research Intern

Research topic: Low-latency and fairness-aware traffic scheduling in distributed block storage systems.

Publications

* *indicates lead author / project lead and corresponding author.*

1. Xiyuan Liang, **Wenzhe Zhu***, Chang Feng, Weiqiang Yu, Xinran Zhang, and Weifa Liang. "XORbit: Bypassing Update and Repair Penalties in Erasure-Coded KV Stores via Dynamic Decoupling." To appear in *International Symposium on Reliable Distributed Systems* (**SRDS 2026**).
2. Haoting Tang, **Wenzhe Zhu***, Qingyang Zhang, Jiakun Zhang, Junlin Jiang, Zaigui Zhang, Haijun Zhang, Yongkun Li, and Yinlong Xu. "CoCache: Accelerating Reads in KV Stores via Cooperative Metadata and Data Cache Management." To appear in *ACM International Conference on Supercomputing* (**ICS 2026**).
3. Jiakun Zhang, Patrick P. C. Lee, **Wenzhe Zhu**, Yongkun Li, Shuyi Zhang, and Yinlong Xu. "Cooperative Scheduling Between CPU and Disk I/O for Load Balancing in Distributed Key-Value Stores." *IEEE International Conference on Data Engineering* (**ICDE 2026**).
4. Zhiqiang Wang, **Wenzhe Zhu***, Zaigui Zhang, Chaomei Yan, Fan Guo, Yongkun Li, and Yinlong Xu. "Amber: Towards Fast and Space-Efficient Incremental Checkpointing in Large Language Model Training." *International Conference on Parallel Processing* (**ICPP 2025**).
5. Shuai Lin, Rui Wang, Zaigui Zhang, Long Deng, **Wenzhe Zhu**, Yongkun Li, and Yinlong Xu. "PTWalker: Cache-Efficient Random Walks via Alternating Dual-Subgraph Walker Updating." *International Conference on Parallel Processing* (**ICPP 2025**).
6. Xiaoyang Wang, Yongkun Li, Kan Wu, **Wenzhe Zhu**, Yuqi Li, and Yinlong Xu. "FineMem: Breaking the Allocation Overhead vs. Memory Waste Dilemma in Fine-Grained Disaggregated Memory Management." *USENIX Symposium on Operating Systems Design and Implementation* (**OSDI 2025**).

7. Tao Li, Yongkun Li, **Wenzhe Zhu**, Yinlong Xu, and John C. S. Lui. "MinFlow: High-Performance and Cost-Efficient Data Passing for I/O-Intensive Stateful Serverless Analytics." *USENIX Conference on File and Storage Technologies* (**FAST 2024**).
8. **Wenzhe Zhu**, Yongkun Li, Erci Xu, Fei Li, Yinlong Xu, and John C. S. Lui. "DiffForward: On Balancing Forwarding Traffic for Modern Cloud Block Services via Differentiated Forwarding." *ACM SIGMETRICS International Conference on Measurement and Modeling of Computer Systems* (**SIGMETRICS 2023**).
9. Haoyu Mao, Yongkun Li, **Wenzhe Zhu**, Fei Li, and Yinlong Xu. "On Optimizing Traffic Imbalance in Large-Scale Block-Based Cloud Storage: Trace Analysis and Algorithm Design." *IEEE International Conference on Parallel and Distributed Systems* (**ICPADS 2022**).

Future Research Directions

Going forward, I aim to design storage-centric system support for LLM-based agents deployed across end devices and edge servers, improving service quality and resource efficiency under tight memory, storage, compute, and latency constraints. This direction connects my background in storage and memory systems with emerging edge AI workloads, where efficient data management is critical for supporting retrieval, reasoning, and memory operations close to users. In particular, I plan to explore the following directions:

- **Caching for key-value storage and vector indexes:** designing workload-aware caching mechanisms for key-value stores and persistent vector indexes, with the goal of improving lookup and retrieval latency, reducing memory overhead, and supporting mixed metadata/vector access patterns in AI services.
- **Storage-aware agent serving:** coordinating retrieval, context construction, caching, and inference-side data access across end devices, edge servers, and remote servers to balance latency, resource efficiency, and answer quality in agent services.

Technical Skills

- **Systems:** Distributed storage, key-value stores, vector indexing, disaggregated memory, RDMA, Linux kernel.
- **Programming:** C/C++, Python, Go, Shell scripting.